

Revised ePC Experimental Programme: Homotopy Absorption, Fibre Emergence, and the Deployment Settle

Working plan v1.0 — for implementation by coding agents

Causal Fibres / Hyperon ePC effort

July 23, 2026

Abstract

This document restructures the ePC / causal-fibres experimental programme in light of two external results due to Mesto (2026): (i) homotopy distillation of GPT-2 (124M) into an error-parameterised predictive-coding (ePC) network, near-losslessly and reproducibly, via a backpropagation-anchored deformation schedule; and (ii) a companion demonstration that the same transformer’s forward pass runs natively inside the MORK symbolic store, with a strongly rank-concentrated PC error field (top 5% of error cells carrying $\approx 90\%$ of squared update mass). Three structural changes follow for our programme: the direct-absorption stage previously abandoned after the R8/R9 collapse is replaced by a guarded homotopy variant and re-run first (E1); frontier-stability measurement is promoted to the keystone experiment deciding the emergence question (E2); and MORK integration, previously a long-horizon item, is pulled forward (E5). Two further experiments test whether fibre structure earns its keep over raw rank concentration (E3) and whether any teacher-free settle objective closes the amortization gap (E4). All experiments run on the six-layer synthetic-grammar rig, where ground-truth generative factors and genuine do-operations are available. This document specifies, for each experiment: hypothesis, setup, conditions, metrics with formulas, pre-registered acceptance criteria, branch logic, and implementation notes.

Contents

1 Context: what changed and why the plan is being revised

1.1 Prior internal results assumed known

The reader (human or agent) is assumed to have access to the causal-fibres synthesis report and sandbox codebase. The facts this plan builds on:

1. **R8/R9 (absorption collapse).** Direct ePC distillation of the six-layer GPT-2-style teacher under energy-monotonicity training concentrated update credit in the final block and collapsed representation rank. This motivated the frozen-backbone + sidecar architecture of the causal-fibres design.
2. **S2 (energy descent $\neq$ causal correctness).** Energy minimisation can coexist with causally wrong credit assignment; correctness must be established by interventional diag-

nostics, not by energy curves.

3. **S3/S4 (supplied-structure successes).** With supplied supports and supplied fibres in canonical coordinates, fibre-local plasticity gives zero unrelated drift, and transports must be identified separately from routing. All positive sandbox evidence lives in this supplied-structure regime; nothing yet tests *emergence* of fibre structure from a trained model.
4. **v0.3 JBD.** The robust joint block diagonalizer recovers hidden common block structure from context-conditioned operators and flags repeated-copy underidentification.

1.2 External input: Mesto (2026)

Two results, treated here as provisional but load-bearing:

1. **Homotopy distillation at 124M.** A pretrained GPT-2 (124M) is deformed into an ePC network along a schedule $\tau = \eta T$, $T \in \{1, 2, 4, \dots, 128\}$, anchored at the exact BP identity at $T=1$ (at $\varepsilon = 0$ the forward pass is the vanilla network and one relaxation step reproduces the backpropagation adjoint). Across five runs, student–teacher prompt KL stays near 10^{-5} nats at every milestone; held-out perplexity within 0.4%; per-block cosine to teacher ≥ 0.998 ; while the mean per-block error-norm ratio rises $0.05 \rightarrow 0.98$.
2. **Error-mass concentration.** At the terminal settle, the top 5% of the 4.72M error cells carry $\approx 90\%$ of squared update mass; fixed-depth Hill statistic $\alpha \approx 1.68$ at the top-5% tail, nearly τ -invariant; no power-law plateau under tail-depth sweep.
3. **MORK companion.** The full GPT-2 (124M) forward pass runs natively in the MORK trie-backed store to 6.06×10^{-6} maximum relative logit error, with KV-caching derived as incremental view maintenance and a blocked tensor format giving a $16\times$ expression-count reduction.

Known open ambiguities in Mesto’s report (communicated to the author; our own experiments must not inherit them):

- *Clamped vs. deployed.* The settle energy’s loss term is the distillation KL to the frozen teacher, so all reported fidelity numbers are measured with teacher access during settle. The teacher-free $\varepsilon=0$ feedforward fidelity of the terminal- τ student is unreported. Every fidelity metric in this plan is therefore reported in *both* clamped and teacher-free conditions, always.
- *Depth distribution of the frontier.* Concentration is reported over cells but not over blocks; whether the top- k frontier piles into late layers (the R8/R9 pathology in slow motion) is unknown. Every frontier measurement in this plan is reported per block as well as pooled.

1.3 The three structural changes

1. **Absorption returns, homotopy-guarded (E1).** The frozen-backbone compromise was forced by R8/R9. If a BP-anchored homotopy with fidelity guards avoids the collapse on our rig, absorption stops being the dangerous last stage.
2. **Emergence becomes the keystone (E2).** Mesto’s concentration result supplies the raw material (a small update frontier) but not the structure (stable, context-indexed subspaces). Whether fibres *emerge* from a homotopied error field, under the most favourable conditions

that will ever exist (few layers, few factors, ground-truth do-operations), decides the programme’s central claim.

3. **MORK moves forward (E5).** The store demonstrably hosts a model $20\times$ larger than ours. Hosting the six-layer PC model in MORK de-risks the long-term target and supplies the substrate in which local credit assignment is an engineering constraint rather than an aesthetic preference.

2 Shared platform, notation, and methodology standards

2.1 The rig

All experiments use the existing six-layer GPT-2-style transformer rig trained on the synthetic compositional grammar with F ground-truth generative factors ($F = 4$ in the current configuration; `TODO(agent)`: read F and factor names from the rig config rather than hard-coding). The rig provides:

- a frozen **teacher** $\mathcal{T}$ (the trained six-layer model);
- a **context-signature** map $c(x)$ assigning each input its ground-truth factor configuration;
- **do-operations** $x \mapsto x^{\text{do}(j)}$ producing counterfactual minimal pairs that intervene on factor j only;
- held-out splits: in-distribution (ID), and **compositional shift** (CS): factor-value combinations excluded from training.

2.2 ePC student parameterisation

Following the Mesto/ePC form. Blocks f_ℓ , $\ell = 1, \dots, L$ ($L=6$), embedding output x_0 , explicit error tensors ε_ℓ as the only free inference leaves:

$$x_\ell = f_\ell(x_{\ell-1}) + \varepsilon_\ell, \quad \hat{z} = \text{head}(\text{LN}_f(x_L)). \quad (1)$$

Inference relaxes

$$E(\varepsilon) = \sum_{\ell=1}^L \frac{1}{2} \|\varepsilon_\ell\|^2 + \mathcal{L}(\hat{z}(\varepsilon)) \quad (2)$$

by T steps of gradient descent from $\varepsilon^{(0)}=0$ with weights frozen (assert at runtime). Weights then descend the local objective

$$E_W = \sum_{\ell=1}^L \frac{1}{2} \|f_\ell(\bar{x}_{\ell-1}; W_\ell) - \bar{x}_\ell\|^2 + \mathcal{L}(\text{head}(\text{LN}_f(\bar{x}_L))), \quad (3)$$

with settled states $\bar{x}$ held constant; optimiser AdamW. The loss term $\mathcal{L}$ varies by experiment: distillation KL to $\mathcal{T}$ (E1, E2, E3), observed-token cross-entropy or symbolic-constraint energy (E4).

2.3 Homotopy schedule and guards (shared by E1–E3)

- **Schedule:** $T \in \{1, 2, 4, \dots, T_{\max}\}$ geometric; effective relaxation time $\tau = \eta T$, η fixed within a run; start $T_{\max}$ at 128 and reduce if the rig converges earlier.
- **Stage advance** requires an energy-convergence hold criterion (energy-convergence ratio below threshold for H consecutive evaluations); on failure, subdivide the τ step.
- **Divergence-abort guard:** abort a run if clamped prompt KL to teacher exceeds a threshold fixed in advance. Log guard status every milestone; report whether the guard ever fired.

2.4 Methodology standards (apply to every experiment)

These adopt the replicate-campaign lesson from Mesto §4 and our own S-series practice. They are **mandatory**; agents must implement them as shared harness code, not per-experiment ad hoc scripts.

1. **No thresholds from single runs.** Acceptance thresholds are set as the strict envelope over a replicate set of $n \geq 3$ runs, under a rule fixed before the final replicates are inspected.
2. **Matched seeds for comparisons.** Any A-vs-B configuration comparison runs both arms at identical seeds and is judged on per-seed deltas.
3. **Pre-registration.** Each experiment’s acceptance criteria (Appendix ??) are frozen in the repo (versioned JSON) before the first full run; deviations are logged, not silently applied.
4. **Dual-condition fidelity.** Every fidelity metric is reported both clamped (settle includes teacher loss) and teacher-free ($\varepsilon=0$ feedforward, no settle, no teacher anywhere in the evaluation path). The harness must make the teacher-free path a first-class evaluator, asserted to touch no teacher weights.
5. **Per-block reporting.** Any statistic over error cells is emitted both pooled and per block ℓ .
6. **Interventional validation.** Any claim of causal credit correctness is validated against do-operation ground truth, never against energy descent (S2).
7. **Artifacts.** Every run emits: config hash, seed, git commit, per-milestone metrics (JSONL), stored error-field histograms, and the standard figure set (§??).

3 E1: Homotopy re-expression of the six-layer teacher (H-R8)

3.1 Question and hypothesis

Does a BP-anchored homotopy with fidelity guards avoid the R8/R9 pathology (final-block credit concentration and representation-rank collapse) that direct energy-monotonic ePC distillation produced on the same rig?

Hypothesis 1. *Under the homotopy schedule, the six-layer student reaches terminal τ with (i) teacher-free $\varepsilon=0$ fidelity within the acceptance envelope, (ii) a depth-distributed update frontier, and (iii) no representation-rank collapse relative to the teacher.*

3.2 Setup

- Teacher: frozen six-layer rig model. Student: initialised from teacher, ePC parameterisation, $\mathcal{L} = \text{KL}(\text{student} \parallel \text{teacher})$ at distillation temperature 2.
- Homotopy and guards as in §2. Replicates: $n \geq 5$ (2 seeds $\times$ 2 microbatch shapes + pilot), matching Mesto’s design so results are directly comparable.
- **Contrast arm (required)**: re-run the original R8/R9 recipe (direct energy-monotonic training, no homotopy anchor, no guards) at matched seeds, same token budget, as the pathology-positive control. If the historical run is reproducible from the archived config, reuse it; otherwise rerun.

3.3 Diagnostics

D1 (teacher-free fidelity).

At every milestone and at terminal τ : prompt KL and held-out task metric for (a) clamped settle, (b) $\varepsilon=0$ feedforward. Primary result is (b); (a) is the sanity ceiling.

D2 (frontier depth histogram).

At terminal τ : fraction of top- k update-mass cells per block, k at the 5% depth, plus the full per-block mass shares. Compare against the contrast arm’s histogram.

D3 (representation rank).

Per block: effective rank (participation ratio of the activation covariance spectrum, $\text{PR} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$) of settled states, tracked across τ , compared to teacher and to the contrast arm.

D4 (concentration curve).

Cumulative squared-update-mass vs. rank-ordered cell fraction (Mesto Fig. 3 analogue), pooled and per block; fixed-depth Hill statistic at top-5% with model-free sampling error, plus the tail-depth sweep. Purpose: establish whether the 124M concentration phenomenology reproduces at six layers, which calibrates how far the rig generalises.

3.4 Acceptance and branches

Acceptance thresholds per §2 standards; provisional targets in Appendix ??.

Branch 3.1 (E1-A: homotopy clean). D1(b) within envelope, D2 depth-distributed (no block holds $> 50\%$ of top- k mass; **TODO**: freeze exact bar pre-run), D3 no collapse. Consequence: the frozen-backbone compromise is dissolved for this rig; E2 and E3 run on the homotopied whole model; sidecar variants become ablations rather than the primary architecture.

Branch 3.2 (E1-B: pathology in slow motion). D1(b) passes but D2/D3 show late-block concentration or rank decay trending with τ . Consequence: absorption remains unsafe; the sidecar architecture is *necessary*, not merely cautious; E2 runs on the sidecar error field instead; publish the negative result with the contrast arm.

Branch 3.3 (E1-C: clamped-only fidelity). D1(a) passes but D1(b) fails badly. Consequence: function is held jointly by weights + teacher-clamped settle; E4 (deployment settle) is promoted to run immediately after E1, since nothing downstream is deployable without it; report the finding to Mesto as the resolution of the ambiguity flagged in §1.2.

4 E2: Frontier stability and fibre emergence (keystone)

4.1 Question

All prior positive fibre results are supplied-structure. E2 asks, on the terminal E1 student (or E1-B sidecar): does the settled error field contain *emergent* fibre structure — context-stable supports, common invariant subspaces recoverable by JBD, aligned with the grammar’s ground-truth factors? This is the emergence question posed under the most favourable conditions that will ever exist; a negative here bounds the programme’s ambitions honestly.

4.2 Measurements

All computed on settled error fields over a held-out probe set with balanced context signatures, with matched-size random-cell and shuffled-context nulls for every statistic.

M1 (support stability).

For each context signature c , the mean top- k support set S_c (cells in the top- k by squared update mass, majority-vote across inputs with signature c). Report the within-vs-between contrast

$$\Delta_{\text{Jacc}} = \mathbb{E}_c[\text{Jacc}(S_c^{(1)}, S_c^{(2)})] - \mathbb{E}_{c \neq c'}[\text{Jacc}(S_c, S_{c'})], \quad (4)$$

where $S_c^{(1)}, S_c^{(2)}$ are supports from disjoint halves of the probe set. High within, low between $\Rightarrow$ stable, context-indexed frontier. Report at $k \in \{1\%, 5\%, 20\%\}$ depths, per block and pooled.

M2 (subspace stability).

Replace cell sets by subspaces: per context c and block ℓ , the top- r left singular subspace $U_{c,\ell}$ of the stacked settled error matrix. Report principal angles between split-half estimates of $U_{c,\ell}$ (within) and between $U_{c,\ell}, U_{c',\ell}$ (between), as a function of r .

M3 (JBD recovery).

Form context-conditioned second-moment operators $M_{c,\ell} = \mathbb{E}[\varepsilon_\ell \varepsilon_\ell^\top \mid c]$ and run the v0.3 robust JBD across c per block. Report: recovered block structure, the underidentification flags, and the **factor-alignment score**: mutual information between recovered block activation patterns and ground-truth factor values across the probe set (with permutation-null calibration).

M4 (interventional fingerprints).

For each factor j , the differential error field $\delta_j(x) = \varepsilon(x^{\text{do}(j)}) - \varepsilon(x)$. Report whether δ_j concentrates on the j -aligned recovered block (fingerprint selectivity matrix: fraction of $\|\delta_j\|^2$ landing in block b , for all j, b ; success is a dominant diagonal after optimal matching).

M5 (observability floor).

The existing ε_{obs} diagnostic applied to the E1 substrate: how much of the teacher–student functional gap is linearly readable from the settled error field at all. M1–M4 are interpreted relative to this floor.

4.3 Branches

Branch 4.1 (E2-A: emergence). Δ_{Jacc} and M2 significantly above nulls, M3 recovers $\geq \lceil F/2 \rceil$ factor-aligned blocks, M4 diagonal dominant. Consequence: first positive emergence result; the

recipe is *homotopy first, then identification*; E3 proceeds with the recovered fibres as its structured arm.

Branch 4.2 (E2-B: concentration without structure). Concentration reproduces but stability/JBD/fingerprints sit at null levels. Consequence: the “red regime” is real even under ideal conditions; the programme’s honest scope contracts to supplied-structure applications (engineered fibres for engineered subsystems); E3’s structured arm uses supplied fibres and is reframed as measuring the cost of imposing structure.

Branch 4.3 (E2-C: blocked by observability). M5 shows the error field barely encodes the functional gap. Consequence: the substrate, not the identification machinery, is the binding constraint; revisit E1 configuration (deeper relaxation, alternative loss placement) before concluding anything about emergence.

5 E3: Structured vs. unstructured frontier settle at matched budget

5.1 Question

Mesto’s proposed rank-frontier settle is our routing apparatus with the structure deleted: an unstructured top- k gate. Does fibre structure buy anything over raw concentration — in settle cost, in-distribution quality, or (the real bet) out-of-distribution transfer?

5.2 Conditions (all at matched settle evaluation budget)

1. **Dense**: full settle, T steps, all cells (reference).
2. **Top- k** : settle updates only the top- k cells by running magnitude (Mesto-style rank frontier; k swept over $\{1\%, 5\%, 20\%\}$).
3. **Fibre-projected**: settle updates projected onto the E2 fibres (recovered if E2-A, supplied if E2-B), with transports applied across blocks; plasticity gates from the S3/S4 machinery active.
4. **Random- k** (control): settle on a random cell set of matched size.

“Matched budget” means equal total scalar update counts per token; the harness must meter and log this, not assume it.

5.3 Evaluations

V1 (ID quality).

Teacher-free and clamped fidelity on the ID split. Expected: dense $\approx$ top- k $\approx$ fibre $\gg$ random; a tie here is the predicted outcome, not a failure.

V2 (compositional shift).

Same metrics on the CS split (held-out factor combinations). This is the discriminating condition: structured routing should degrade more gracefully than unstructured top- k if fibre structure has content.

V3 (transfer of routing).

Freeze each condition’s routing/gating state on ID data; evaluate settle quality on CS data

without re-estimating the frontier. Measures whether the routing function itself generalises — the evidence previously missing from the ladder.

V4 (continual probe).

Sequential fine-tuning on factor-restricted streams; measure off-support drift per condition. Prediction from S3/S4: protection tracks gating/supports, not PC locality per se; top- k without supports drifts.

6 E4: Deployment settle objectives (closing the amortization gap)

6.1 Question

The teacher-clamped settle cannot be the deployed object. At deployment, $\mathcal{L}$ in Eq. (2) must be replaced by something available at test time, or the settle must be dropped ($\varepsilon=0$). Does *any* teacher-free settle objective beat the feedforward student? If not, the PC apparatus has no deployment value beyond what an adapter provides, and the programme’s deployment story must say so.

6.2 Conditions

On the terminal E1 student, evaluated teacher-free throughout:

1. **FF**: $\varepsilon=0$ feedforward (lower anchor).
2. **TC**: teacher-clamped settle (upper anchor; measures the gap available in principle, never deployable).
3. **OT**: observed-token settle: $\mathcal{L}$ is cross-entropy on the observed prefix tokens (self-supervised settle on the prompt), free continuation.
4. **SC**: symbolic-constraint settle: grammar rules expressed in MeTTa; $\mathcal{L} = \lambda \cdot (\text{constraint-violation energy from } t \text{ queried per settle step (batched; see E5 for the in-store path)})$. This is the Stage-4 bet made concrete: test-time information injected by the symbolic cap.
5. **OT+SC**: combined.

6.3 Evaluation and the decisive comparison

Primary metric: task accuracy / next-token quality on ID and CS splits, plus constraint-satisfaction rate of generated continuations. The decisive number is

$$G = \frac{\text{SC} - \text{FF}}{\text{TC} - \text{FF}} \quad (5)$$

per metric: the fraction of the in-principle settle gap recovered by symbolic clamping. The grammar rules are literally the generative truth of the rig; if $G \approx 0$ here, symbolic clamping will not work anywhere easier, the amortization gap is fatal to the deployment story, and downstream design should treat the PC form as a training-time device only. If G is material on CS in particular (where FF should be weakest), the Stage-4 inversion is vindicated: the symbolic cap is where deployment value enters, and it should move up the staging accordingly.

Secondary: settle-step count to criterion per condition (cost of the recovered quality), and interaction with E3 conditions (does fibre-projected SC settle localise constraint corrections to the right blocks? cross-reference M4).

7 E5: MORK hosting of the six-layer PC model

7.1 Question and scope

Mesto’s companion result hosts a 124M forward pass in MORK at 6×10^{-6} relative error with incremental decoding. Our model is $\sim 20\times$ smaller; hosting it, *including the error field and frontier writes*, de-risks the store-native target and supplies the substrate in which local credit is an engineering constraint. Scope for this stage: inference + settle + frontier weight-writes in-store; from-scratch in-store training is out of scope.

7.2 Work items

1. **W1 (forward)**. Six-layer forward as MORK rule application over blocked tensor expressions, dense kernels (einsum, attention, normalisation) as sink operations, following Mesto’s blocked format. Fidelity target: match the PyTorch reference to $\leq 10^{-5}$ maximum relative logit error; exact argmax agreement on a decode probe set.
2. **W2 (settle in-store)**. The T -step relaxation as in-store iteration; error tensors as store expressions. Verify settled states match the reference settle to a pre-registered tolerance.
3. **W3 (frontier writes as IVM)**. Weight updates restricted to the top- k frontier, applied as incremental store writes; measure write volume vs. dense updates and verify the store’s incremental maintenance reproduces the reference update to tolerance. This is the concrete synergy claim (sparse frontier $\leftrightarrow$ trie-backed incremental writes) tested rather than asserted.
4. **W4 (fibre registration)**. Register E2 fibres (if any) as MeTTa expressions: supports, subspace bases, transports, factor bindings. This is the substrate E4’s SC condition queries, and the interface the symbolic cap was always intended to have.
5. **W5 (cost accounting)**. Wall-clock and expression-count budgets for W1–W3, compared against the PyTorch path, reported without spin: the store will be slower; the question is by how much and where.

8 Controls and baselines (run alongside, not after)

C1 (BP-residual observability control).

The standing falsifier for the interface: a plain backprop-trained residual/adaptor of matched capacity on the frozen teacher, evaluated on every E3/E4 metric. If C1 matches the PC apparatus everywhere teacher-free, the parameterisation is doing no deployment work and the reports must say so.

C2 (dictionary-learning baseline).

Sparse autoencoder / dictionary learning trained directly on settled error cells (per block), same probe set as E2. Evaluate on M3’s factor-alignment score and M4’s fingerprint selectivity. If sparse coding recovers the factors as well as JBD, the simpler method wins and

the JBD machinery is repositioned as a verification tool. This baseline is mandatory in any write-up of E2.

C3 (matched-compute BP fine-tune).

For any continual-learning claim (E3 V4): backprop fine-tuning at matched compute and matched seeds, per-seed deltas.

9 Sequencing and decision tree

Phase 1 (immediate, parallelisable): E1 (with its contrast arm) and the harness work in §2. C1 training can start immediately (it depends only on the frozen teacher).

Phase 2: E2 on the Phase-1 substrate (choice of substrate per E1 branch). C2 runs on the same stored error fields — the harness must persist settled error tensors for the probe set precisely so that E2 and C2 consume identical data.

Phase 3: E3 and E4 in parallel; both consume E2’s output (fibres or the finding that there are none). Exception: under branch E1-C, E4 jumps to Phase 2.

Phase 4: E5, overlappable with Phase 3 for W1–W2 (which need only the E1 student).

Reporting rule: each phase produces a short memo in the N-series style — established results, boundaries stated with the same force, branch taken and why — before the next phase’s configs are frozen.

10 Implementation notes for coding agents

- **Repo layout.** Extend the existing causal-fibres sandbox repo: `experiments/e1_homotopy/`, `e2_emergence/`, `e3_settle/`, `e4_deployment/`, `e5_mork/`, `controls/`; shared code in `harness/` (schedules, guards, dual-condition evaluators, metering, metrics JSONL, figure emitters). `TODO(agent)`: reconcile with actual repo structure before creating directories.
- **Config.** One versioned JSON/YAML per run: seeds, schedule, guard thresholds, k depths, probe-set spec, acceptance-criteria file hash. No thresholds in code.
- **Assertions.** (i) weights frozen during settle; (ii) teacher-free evaluators provably teacher-free (no teacher module import in that code path); (iii) matched-budget metering in E3 enforced, not assumed; (iv) matched seeds enforced for all A/B comparisons.
- **Persisted artifacts.** Settled error tensors for the E2/C2 probe set (single source of truth); per-milestone metrics JSONL; error-mass histograms at fixed binning (so Hill statistics and tail sweeps are recomputable without rerunning).
- **Standard figure set.** Per run: fidelity-vs-milestone (clamped and teacher-free, log KL); error-norm-ratio and cosine vs. τ ; concentration curve pooled and per block; frontier depth histogram; effective-rank-vs- τ per block. E2 adds: Δ_{Jacc} vs. k -depth with nulls; principal-angle spectra; JBD block map; fingerprint selectivity matrix. E4 adds: the G bar per metric per split.
- **Numerical hygiene.** Hill statistics computed from stored histograms with the model-free sampling-error estimate; never fit a power law; report the tail-depth sweep alongside any

fixed-depth α .

- **Failure reporting.** A run that trips a guard or fails acceptance is a result, not a discard: it is logged, its artifacts kept, and it appears in the phase memo. No silent exclusion; the n-of-runs reported must equal the n-of-runs launched.

A Provisional acceptance criteria

Provisional targets below; final bars are frozen per the envelope rule (§2) before each experiment’s full campaign, and stored as versioned JSON. Deltas are per-seed unless noted.

Exp.	Quantity	Provisional bar	Notes
E1	clamped prompt KL	within envelope of $n \geq 3$ replicates	guard at fixed multiple
E1	teacher-free task delta	$\leq$ envelope over replicates	primary
E1	frontier depth	no block $> 50\%$ of top-5% mass	freeze before run
E1	effective rank	$\geq 0.8 \times$ teacher PR per block	vs. contrast arm
E2	Δ_{Jacc}	$> 3\sigma$ above shuffled-context null	per k -depth
E2	factor alignment (M3)	$\geq \lceil F/2 \rceil$ blocks above permutation null	with JBD flags clean
E2	fingerprint (M4)	diagonal share $> 2 \times$ off-diagonal mean	after matching
E3	ID tie	top- k , fibre within envelope of dense	tie is expected
E3	CS separation	fibre vs. top- k per-seed delta, sign test	discriminating
E4	G	report; material if > 0.2 on CS	decisive number
E5	W1 fidelity	$\leq 10^{-5}$ max rel. logit error; argmax exact	Mesto parity

B Metric definitions gathered

- **Squared update mass** of cell i : $m_i = \mathbb{E}[\varepsilon_i^2]$ over the probe set at the terminal settle (store per-sample values for split-half work).
- **Top- k support** S_c : cells in the top k fraction of m_i computed on inputs with signature c (majority vote across samples).
- **Jaccard**: $\text{Jacc}(A, B) = |A \cap B| / |A \cup B|$.
- **Effective rank (PR)**: $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ over the eigenvalues of the per-block settled-state covariance.
- **Hill statistic at depth k** : with order statistics $m_{(1)} \geq \dots \geq m_{(k)}$, $\hat{\alpha}^{-1} = \frac{1}{k} \sum_{i=1}^k \ln(m_{(i)}/m_{(k)})$; report with the model-free dispersion-based standard error; descriptor only, not an exponent estimate.
- **Fingerprint selectivity**: matrix $\Phi_{jb} = \mathbb{E}[\|P_b \delta_j\|^2] / \mathbb{E}[\|\delta_j\|^2]$ for do-difference fields δ_j and recovered block projectors P_b ; optimal j - b matching by the Hungarian algorithm before diagonal/off-diagonal comparison.
- **G (gap recovery)**: $(\text{SC} - \text{FF}) / (\text{TC} - \text{FF})$ computed per metric per split; undefined and reported as such when $\text{TC} - \text{FF}$ is within noise.